A PROJECT REPORT
ON

“Hotel FAQ Chatbot”

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CERTIFICATE

This is to certify that the project report entitled **"Hotel FAQ Chatbot"** Was Successfully completed by Student of sixth semester Diploma in (Computer Engineering).

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in partial fulfilment of the requirements for the award of the Diploma in (Computer Engineering) and submitted to the Department of (Computer Engineering) of Government Polytechnic, Jintur work carried out during a period for the academic year 2025-26 as per curriculum.

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Abstract

The “Hotel FAQ Chatbot System” is an intelligent web-based application designed to provide automated responses to frequently asked questions related to hotel services. The main objective of this project is to enhance customer support by offering quick, accurate, and 24/7 assistance without the need for human intervention.

This chatbot is developed using modern technologies and is capable of understanding user queries and providing relevant responses regarding room booking, availability, pricing, check-in and check-out timings, facilities, and other hotel-related information. The system helps reduce the workload on hotel staff while improving customer satisfaction through instant communication.

The chatbot interface is user-friendly and can be easily accessed through a website or application. It ensures efficient handling of multiple user queries simultaneously and minimizes response time. Additionally, the system can be updated with new information, making it scalable and adaptable to different hotel requirements.

In conclusion, the Hotel FAQ Chatbot System serves as a smart solution for automating customer interaction in the hospitality industry, making services more efficient, reliable, and user-friendly.

CHAPTER 1: INTRODUCTION

1.1Overview of the Project

The rapid growth of digital technology has significantly transformed the hospitality industry. Customers now expect instant responses to their queries regarding hotel services, facilities, and availability. To meet these expectations, automation and artificial intelligence (AI) have become essential tools.

The Hotel Chatbot System Using AI is a web-based application designed to provide automated responses to user queries related to hotel services. The system uses Artificial Intelligence (AI) to understand user input and generate intelligent responses based on stored hotel data.

The chatbot interacts with users through a simple chat interface. When a user asks a question (for example, about room availability, services, or location), the system processes the input using an AI model and retrieves relevant information from a structured JSON data file. The response is then displayed to the user in real time.

The system is developed using:

- Python programming language
- Flask web framework
- Groq API for AI processing
- LLaMA 3.1 model for intelligent responses

This system eliminates the need for human intervention in answering repetitive queries and ensures 24/7 availability.

1.2 Problem Statement

In the traditional hotel management system, customer queries are handled manually by staff through phone calls, emails, or in-person interactions. This approach has several limitations:

- Delayed response time
- High operational cost
- Limited availability (not 24/7)
- Human errors in communication
- Difficulty handling multiple users simultaneously

Customers today expect instant and accurate responses. Manual systems fail to meet this demand efficiently.

Therefore, there is a need for an automated intelligent system that can:

- Respond instantly to customer queries
- Provide accurate hotel information
- Reduce human workload
- Operate continuously without interruption

The proposed system addresses these issues by implementing an AI-based chatbot.

1.3Objectives of the Project

The main objectives of the Hotel Chatbot System are:

1. To develop an intelligent chatbot for handling hotel-related queries
2. To provide instant and accurate responses using AI
3. To reduce dependency on human staff for answering queries
4. To utilize structured data (JSON) for information retrieval
5. To improve customer experience through automation
6. To maintain conversation history for better interaction
7. To provide a simple and user-friendly interface

1.4 Scope of the Project

The scope of this project includes:

Answering queries related to:

1. Hotel services
2. Room types
3. Facilities
4. Location details
5. Providing real-time responses
6. Operating without user login or registration
7. Supporting continuous interaction
8. Support multiple languages

Limitations:

1. Does not handle bookings or payments
2. Works only with predefined hotel data

1.5 Methodology

The working methodology of the system is as follows:

1. User enters a query in the chatbot interface
2. The query is sent to the Flask backend
3. The backend forwards the query to the AI model (via Groq API)
4. AI processes the query using LLaMA model
5. Relevant information is extracted from JSON data
6. AI generates a smart response
7. Response is displayed to the user
8. Chat history store by context
9. t history is stored for context

1.6 Applications of the System

- Hotels and resorts

- Customer support automation
- Travel and tourism industry
- Information assistance systems

1.7Advantages of Hotel Chatbot System (Short – 5 Points)

- Provides instant responses to customer queries.
- Reduces workload on hotel staff.
- Ensures 24/7 availability of customer support.
- Improves customer satisfaction and service quality.
- Saves time and operational costs for hotel management.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

A literature review provides an overview of existing technologies and systems related to the project. It helps in understanding the evolution of chatbot systems and the role of AI in modern applications.

2.2 Traditional Chatbots

Traditional chatbots were based on predefined rules and scripts. These systems worked using:

- Keyword matching
- Decision trees
- Static responses

2.3 AI-Based Chatbots

Modern chatbots use Artificial Intelligence and Natural Language Processing (NLP). These systems can:

1. Understand user intent
2. Generate dynamic responses
3. Learn from interactions

Advantages:

- More accurate responses
- Better user experience
- Ability to handle complex queries

2.4 Large Language Models (LLMs)

Large Language Models like LLaMA are advanced AI systems trained on large datasets. They can:

- Understand human language
- Generate meaningful responses
- Provide contextual answers
- Support multiple languages

These models are widely used in:

1. Chatbots
2. Virtual assistants
3. Customer support systems

2.5 Chatbots in Hospitality Industry

In the hotel industry, chatbots are used for:

- Answering customer queries
- Providing service information
- Assisting in bookings
- Enhancing customer engagement

Benefits:

- 24/7 availability
- Reduced workload
- Improved customer satisfaction

2.6 Existing Systems

Existing hotel chatbot systems include:

- Rule-based systems with limited responses
- AI-based systems integrated with booking platforms

However, many systems:

1. Require login
2. Are complex to use
3. Do not utilize structured data effectively

2.7 Proposed System Advantages

The proposed system improves upon existing systems by:

1. Using AI for intelligent responses
2. Using JSON for structured data
3. Providing simple interface
4. Eliminating need for login
5. Supporting continuous conversation

CHAPTER 3: SYSTEM ANALYSIS

3.1 Introduction

System analysis involves understanding the requirements and specifications of the system before development. It includes functional and non-functional requirements.

3.2 Functional Requirements

Functional requirements define what the system should do.

The system must:

1. Accept user queries through chat interface
2. Process queries using AI model
3. Retrieve data from JSON file
4. Generate intelligent responses
5. Display responses to user
6. Maintain chat history
7. Provide option to clear chat
8. Restrict responses to hotel-related queries

3.3 Non-Functional Requirements

Non-functional requirements define system performance and quality.

1. Performance

Fast response time

Minimal delay

2. Availability

System should be available 24/7

3. Usability

Simple and easy-to-use interface

4. Reliability

Accurate and consistent responses

5. Scalability

Can be extended with more features

3.4 Software Requirements

The software required for the system includes:

1. Python (Programming Language)
2. Flask (Web Framework)
3. Groq API (AI Integration)
4. LLaMA Model (AI Processing)
5. VS Code (Development Environment)
6. Web Browser (Chrome, Edge, etc.)

3.5 Hardware Requirements

Minimum hardware requirements:

1. Processor: Intel i3 or above
2. RAM: 4 GB or more
3. Storage: 10 GB free space
4. Internet connection

3.6 Feasibility Study

1. Technical Feasibility

Uses available technologies (Python, Flask, AI APIs)

Easy to implement

2. Economic Feasibility

Low cost (no expensive hardware required)

Uses free/open-source tools

3. Operational Feasibility

Easy for users to interact

No training required

3.7 System Constraints

Requires internet connection

Depends on API availability

Limited to predefined hotel data

CHAPTER 4: SCOPE OF THE PROJECT

4.1 Introduction

The scope of a project defines the boundaries and the extent to which the system can be used. It describes what the system can do and what it cannot do.

The Hotel Chatbot System Using AI is designed to provide automated assistance to users by answering queries related to hotel services and facilities.

4.2 Scope of the System

The system is capable of performing the following functions:

1. Query Handling

Accepts user queries through chat interface

Supports natural language input

2. Information Retrieval

Retrieves hotel-related data from JSON file

Provides details such as:

Room types

Facilities

Services

Location

3. AI-Based Response Generation

Uses AI model to understand query

Generates intelligent and context-aware responses

4. Chat History Management

Stores previous conversation

Helps in better interaction

5. Chat Reset Feature

Allows user to clear chat

Starts fresh conversation

6. Continuous Availability

System is available 24 hours a day

4.3 Applications of the System

Hotels and hospitality industry

Customer support automation

Information systems

Tourism services

4.4 Limitations of the System

1. No booking functionality
2. No payment integration
3. Works only with predefined JSON data
4. Does not support multiple languages
5. Requires internet connection for AI API

CHAPTER 5: SYSTEM DESIGN

5.1 Introduction

System design is the process of defining the architecture, components, and data flow of the system. It provides a blueprint for implementation.

5.2 System Architecture

Description:

The system follows a client-server architecture:

Client Side:

User interacts via web interface

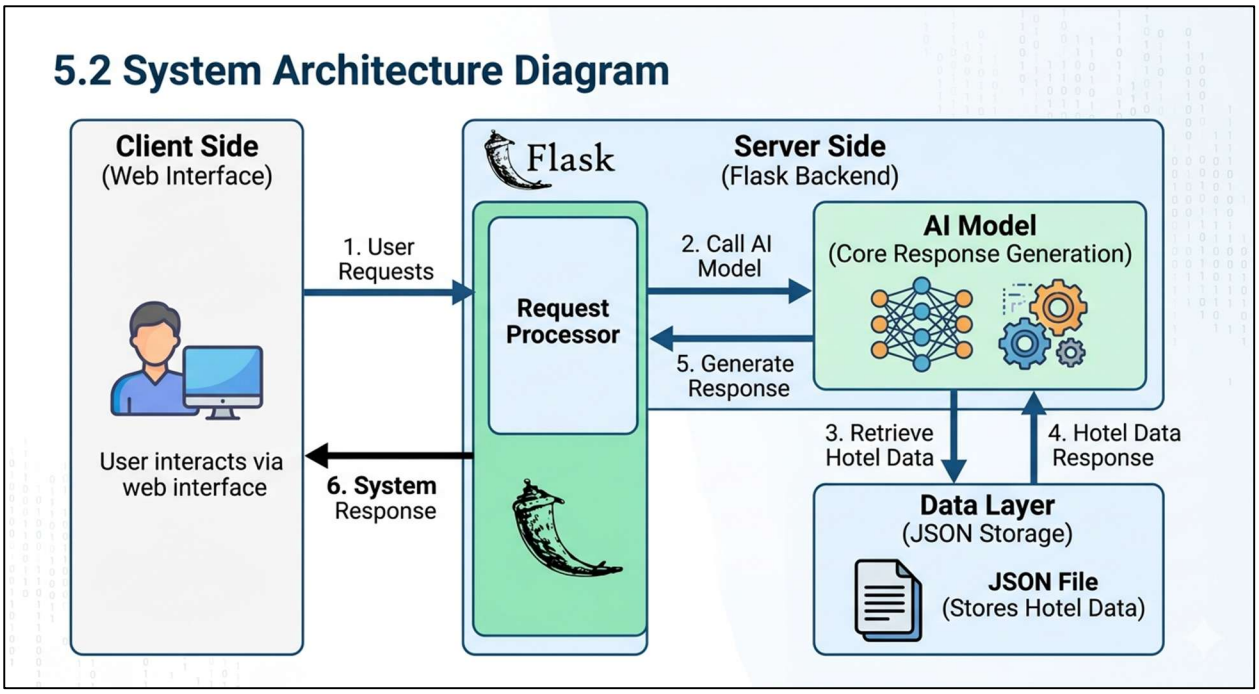
Server Side:

Flask backend processes requests

AI model generates responses

Data Layer:

JSON file stores hotel data



System Architecture Diagram

Diagram no. 1

5.3 Use Case Diagram

Description :

<u>Actor:</u>

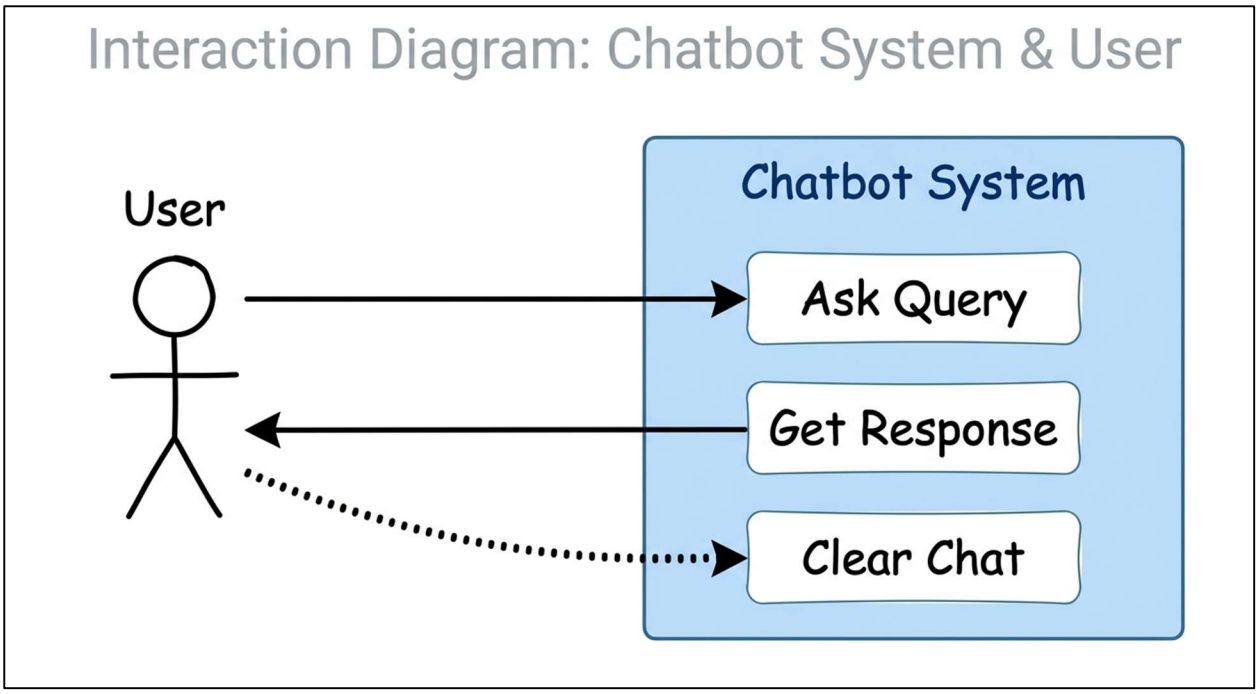
User

<u>Use Cases:</u>

Enter Query

View Response

Clear Chat



Ineraction Diagram:Chatbot System & User

Diagram no. 2

Diagram Explanation: The user interacts with the chatbot system by entering queries. The system processes the query and displays the response. The user can also clear the chat history.

5.4 Data Flow Diagram (DFD)

5.4.1 Level 0 DFD

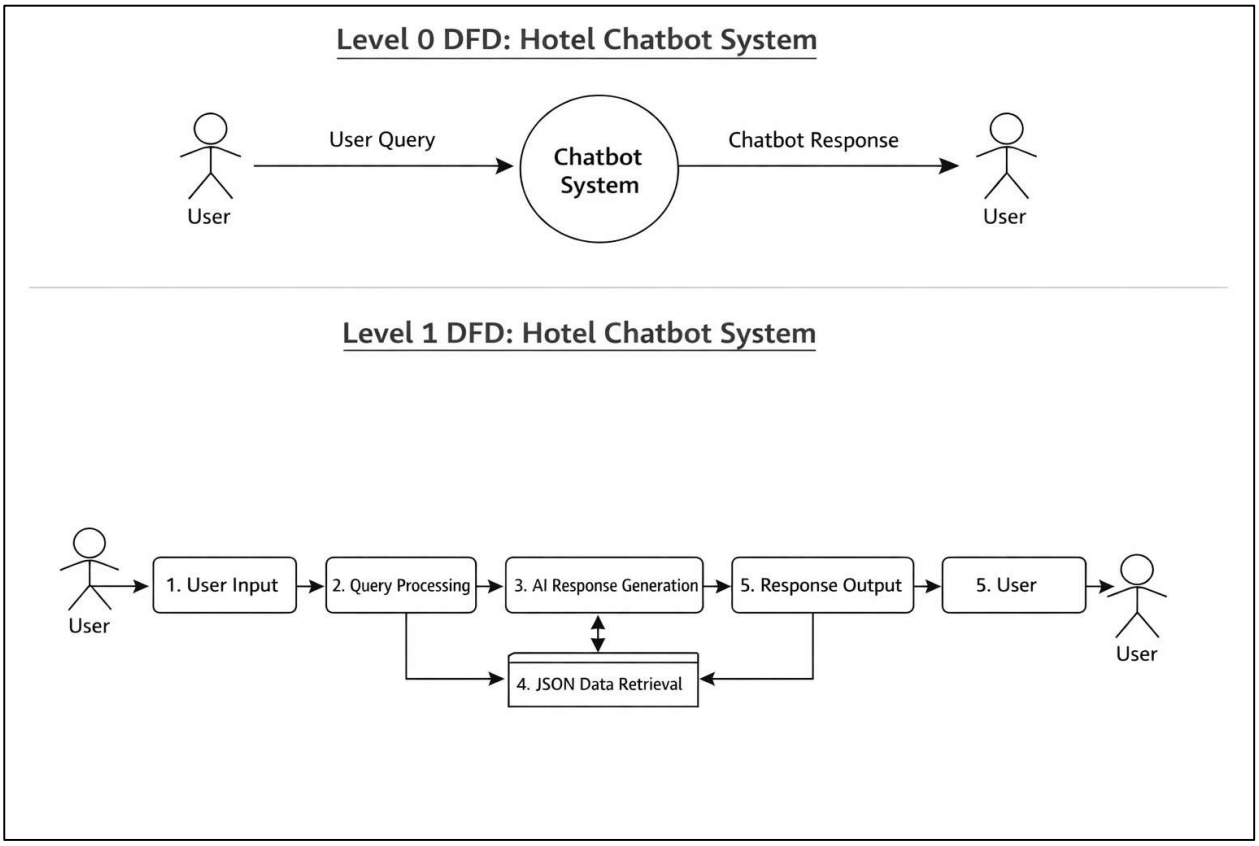
Description: This shows overall system flow.

5.4.2 Level 1 DFD

Components:

- User Input
- Query Processing
- AI Response Generation
- JSON Data Retrieval

- Response Output



DFD Level

Diagram no. 3

5.5 Entity Relationship (ER) Diagram

Entities:

1. User Query
2. Chat History
3. Hotel Data

Relationships:

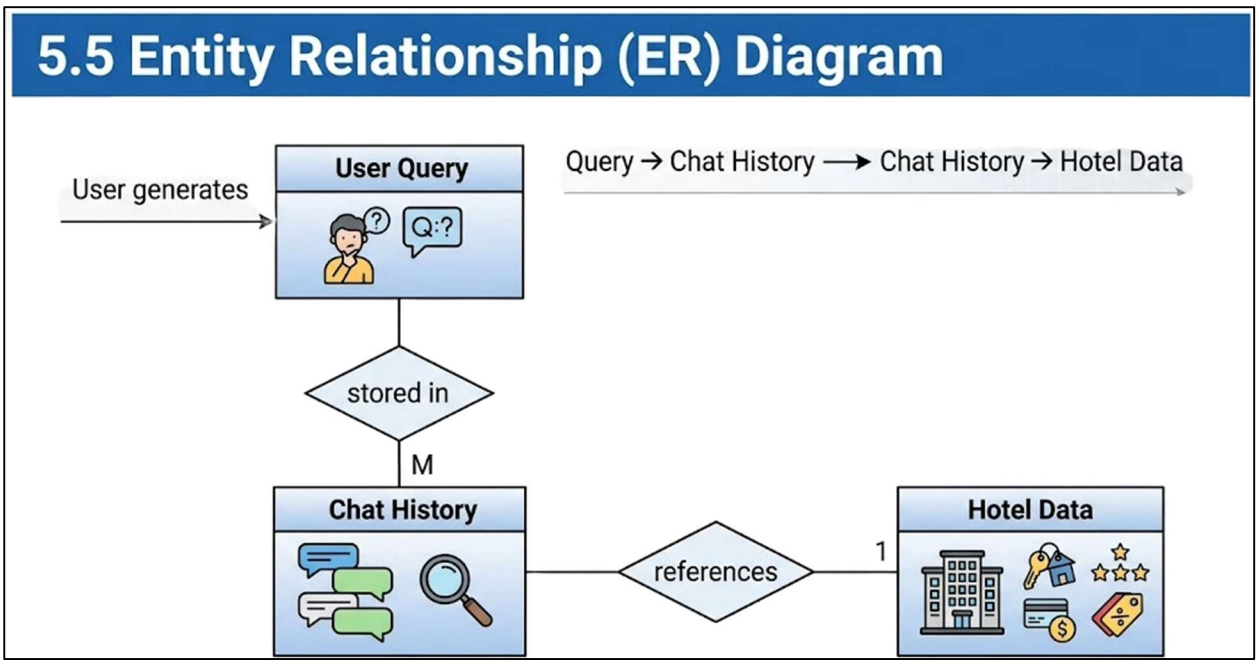
1. User generates Query
2. Query stored in Chat History
3. Chat History references Hotel Data

Diagram Description:

1. Rectangle: User Query
2. Rectangle: Chat History
3. Rectangle: Hotel Data

Connect:

1. Query → Chat History
2. Chat History → Hotel Data



Entity Relational Diagram

Diagram no. 4

5.6 Database Design (JSON Structure):

Since the system uses JSON instead of database, we define structure:

Example JSON Structure:

```
{
  "hotel_name": "Royal Horizon Hotel",
  "location": "Parbhani, Maharashtra",
```

"rooms": ["Standard", "Deluxe", "Suite"],

"facilities": ["Wi-Fi", "AC", "Parking", "Restaurant"],

"check_in": "2:00 PM",

"check_out": "12:00 PM"

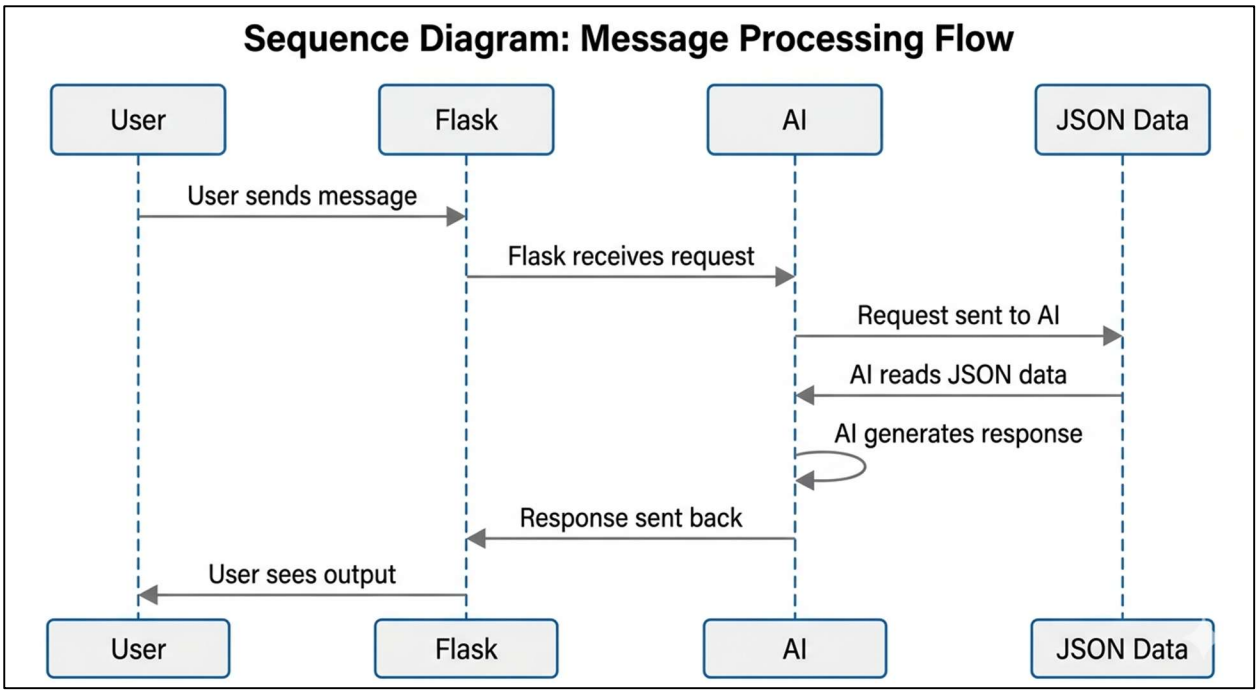
}

5.7 Sequence Diagram

Description: This shows step-by-step interaction.

Flow:

- User sends message
- Flask receives request
- Request sent to AI
- AI reads JSON data
- AI generates response
- Response sent back
- User sees output



Sequence Diagram :Message Prccessing Flow

Diagram no. 5

CHAPTER 6: IMPLEMENTATION

6.1 Introduction

Implementation is the phase where the system is developed using selected technologies. It involves coding, integration, and deployment.

6.2 Development Environment

1. Programming Language: Python
2. Framework: Flask
3. Editor: VS Code
4. API: Groq API
5. AI Model: LLaMA 3.1

6.3 System Modules

6.3.1 User Interface Module

Description:

- Provides chat interface
- Allows user to type queries
- Displays chatbot responses

Features:

- Input text box
- Send button

- Chat display area

6.3.2 Backend Module (Flask)

Description:

- Handles HTTP requests
- Manages routing
- Connects frontend and AI

Example Functions:

- / → Home page
- /chat → Process user query
- /clear → Clear chat

6.3.3 AI Processing Module

Description:

- Sends query to Groq API
- Uses LLaMA model
- Generates smart responses
- Working:
- Receives user input
- Combines with JSON data
- Generates meaningful reply

6.3.4 Data Handling Module

Description:

- Reads JSON file
- Extracts required data
-

Functions:

- Load JSON
- Parse data
- Provide context to AI

6.3.5 Chat History Module

Description:

- Stores previous messages
- Maintains conversation flow

Features:

- Context-aware responses
- Better user experience

6.4 Working Algorithm

Step-by-Step Algorithm:

1. Start system
2. Load JSON data

3. Wait for user input
4. Receive query
5. Send query to AI model
6. Combine query with JSON context
7. Generate response
8. Display response
9. Store in chat history
10. Repeat until user exits

6.5 Key Features Implemented

- No login required
- AI-based responses
- JSON-based data handling
- Chat history support
- Clear chat functionality
- 24/7 availability

6.6 Advantages of Implementation

- Simple architecture
- Easy to maintain
- Low cost
- Scalable

CHAPTER 7: TESTING

7.1 Introduction

Testing is a crucial phase in software development used to verify that the system works correctly and meets the specified requirements. It ensures that the system is free from errors and performs efficiently under different conditions.

7.2 Types of Testing Performed

1. Unit Testing

Individual modules are tested separately

Example: Testing JSON data loading

2. Integration Testing

Testing interaction between modules

Example: Flask + AI API integration

3. System Testing

Entire system is tested as a whole

4. User Acceptance Testing (UAT)

Testing from user perspective

Ensures system meets user needs

7.3 Test Case Design

Test Case Table:

Test ID	Test Description	Input / Action	Expected Output	Actual Output	Status
TC01	Check hotel information (core functionality)	"What facilities do you provide?"	System displays hotel facilities from JSON data	Correct response shown	Pass
TC02	AI-based query handling	"Tell me about deluxe rooms"	AI generates meaningful and detailed response	Correct response shown	Pass
TC03	Unknown query handling	"Do you have a swimming pool?" (not in data)	AI gives smart fallback or approximate answer	Correct response shown	Pass
TC04	Clear chat functionality	Click "Clear Chat" button	Chat history is cleared and reset	Chat cleared successfully	Pass

Table no. 1

7.4 Test Results

- All test cases executed successfully
- System performed as expected
- No major errors found

7.5 Error Handling

The system handles errors such as:

- Invalid input
- Empty queries
- API failures

Appropriate fallback responses are provided to ensure a smooth user experience.

CHAPTER 8: RESULTS AND DISCUSSION

8.1 Introduction

This chapter presents the output of the developed system and discusses its performance.

8.2 System Output

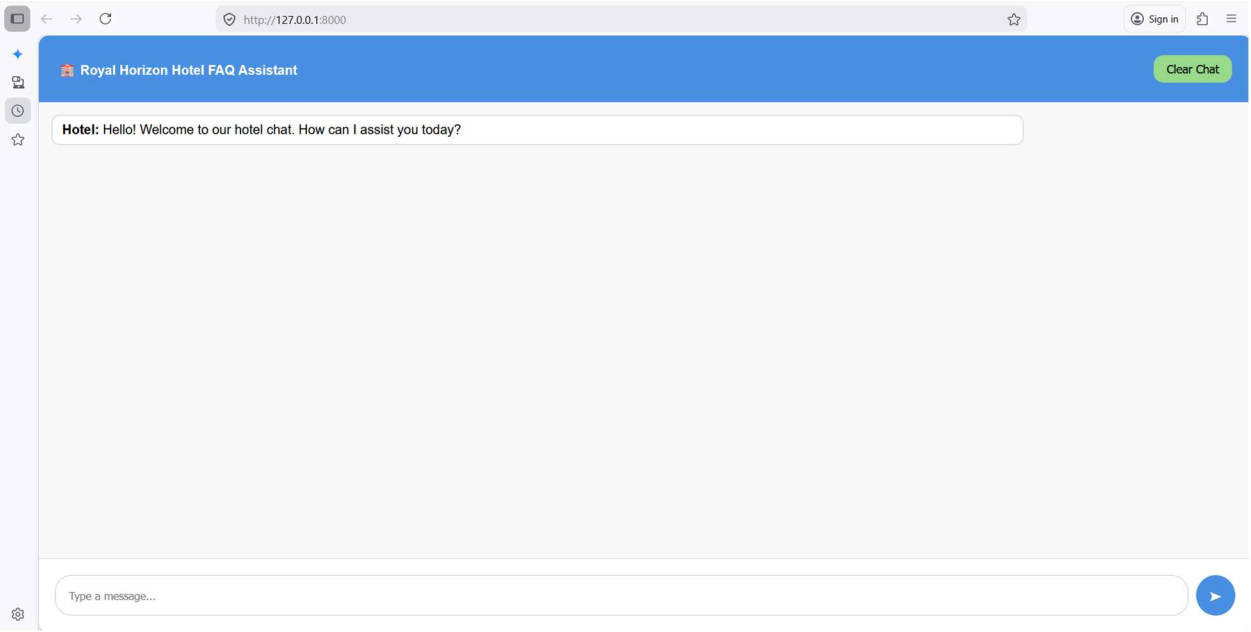
The system successfully performs the following:

- Accepts user queries
- Processes queries using AI
- Retrieves data from JSON
- Displays intelligent responses

8.3 Screenshots

Screenshot 1: Home Page

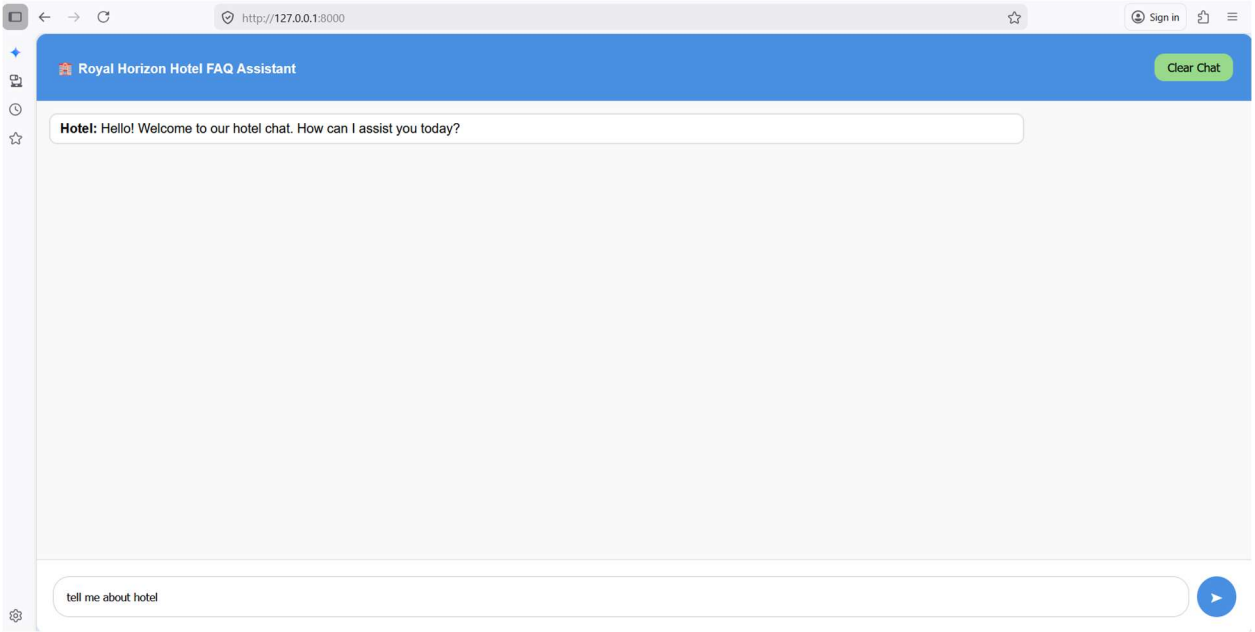
Figure 8.1: Chatbot Home Interface



Screenshot no. 1

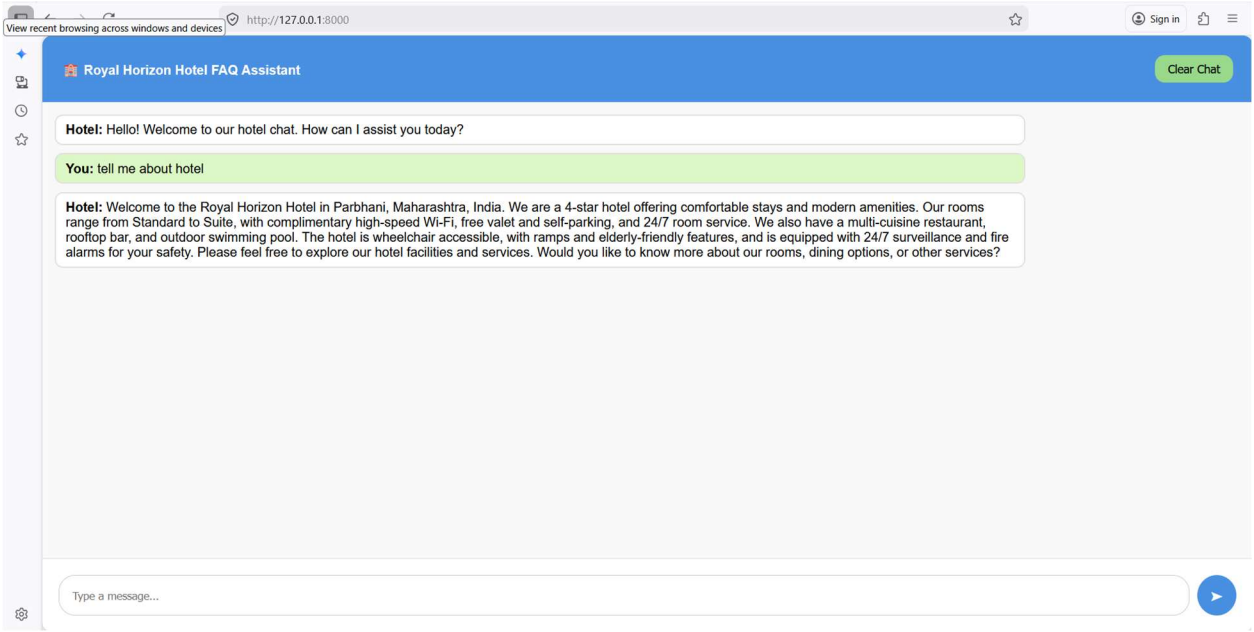
Screenshot 2: All functions Home Page, User Query, AI Response, Chat History, Clear Chat Feature:

1. Home page:

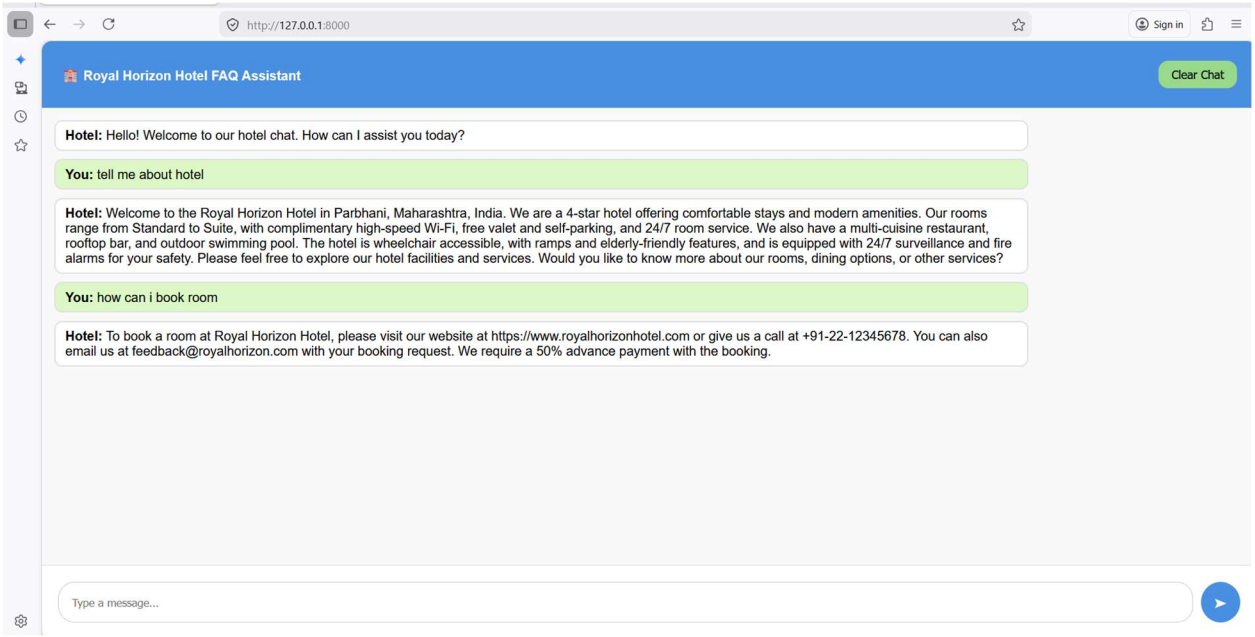


Screenshot no. 2

2. User query and chatbot responce:

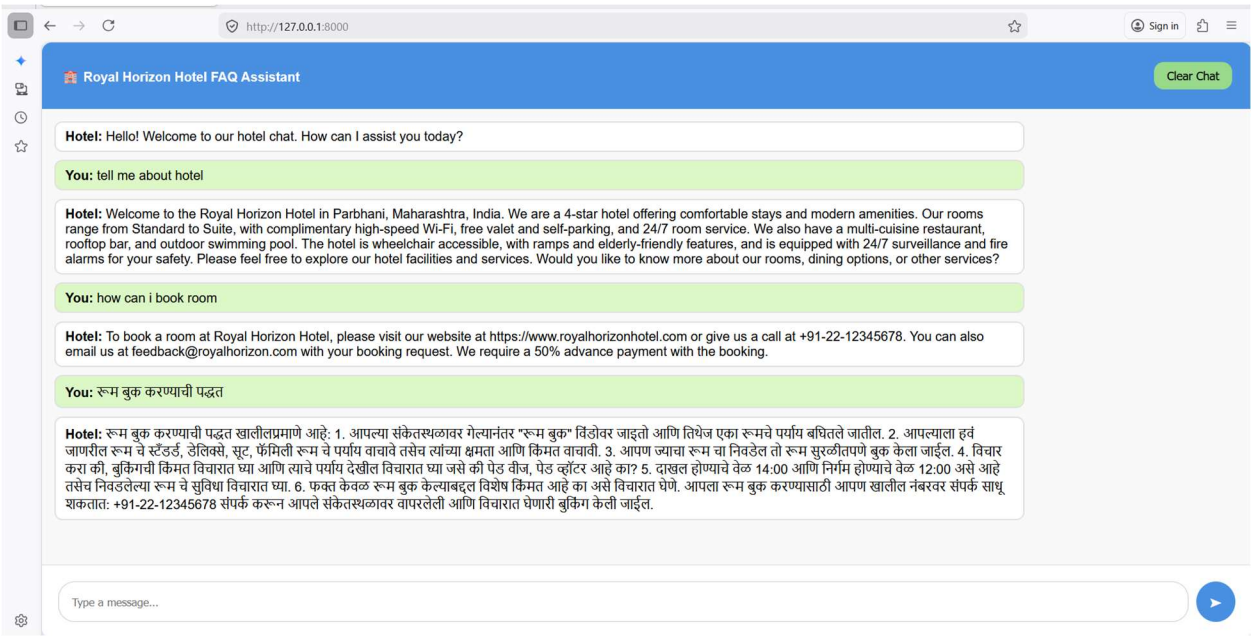


Screenshot no. 3



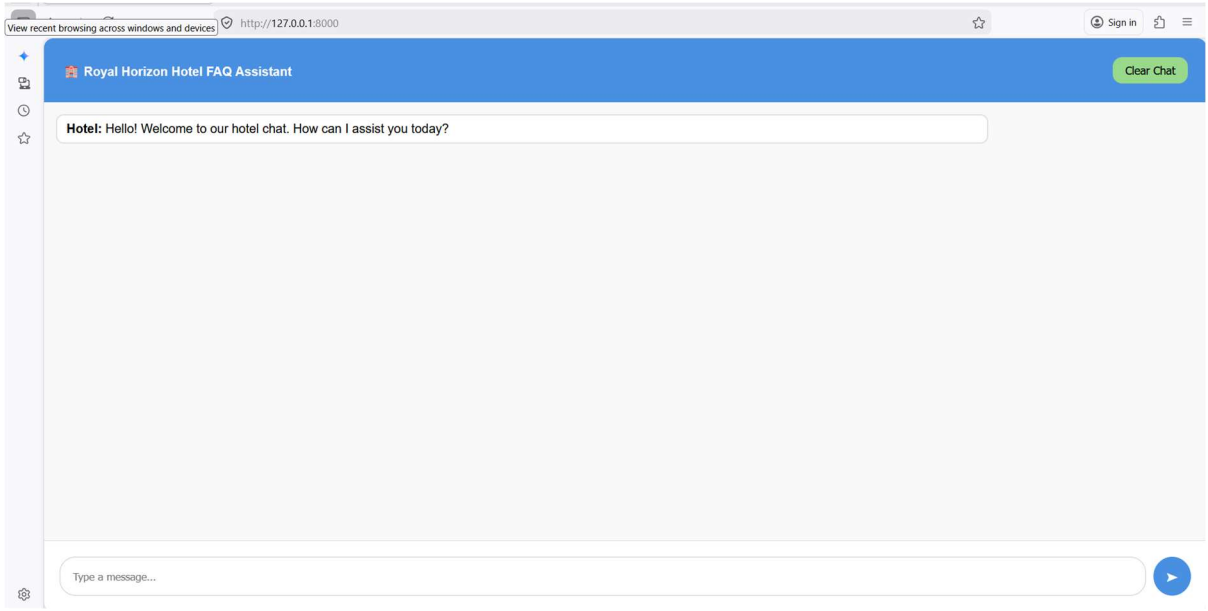
Screenshot no. 4

3. <u>Multiple language support:</u>



Screenshot no. 5

4. <u>Clear chat history:</u>



Screenshot no. 6

8.4 Performance Analysis

- Response time is fast
- AI responses are accurate
- System handles multiple queries efficiently

8.5 Discussion

The chatbot system provides an efficient solution for hotel query handling. The use of AI improves response quality compared to traditional rule-based systems.

Unlike manual customer service, the chatbot can handle multiple user queries simultaneously without delays, making it highly suitable for modern hotel environments.

The system demonstrates how automation can enhance customer satisfaction by providing instant responses to frequently asked questions such as room availability, pricing, facilities, and booking information. This reduces the workload on hotel staff and allows them to focus on more complex customer needs. Additionally, the chatbot operates 24/7, ensuring continuous customer support even outside business hours.

CHAPTER 9: CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The Hotel Chatbot System Using AI has been successfully developed and implemented. The system meets all the objectives defined at the beginning of the project.

Key achievements:

- Developed AI-based chatbot
- Provided real-time responses
- Reduced need for human interaction
- Implemented JSON-based data handling
- Achieved 24/7 availability

The system demonstrates how AI can be effectively used in the hospitality industry to improve customer service.

9.2 Future Scope

The system can be enhanced with the following features:

1. Online Booking System

Allow users to book rooms

2. Payment Integration

Support online payments

3. Voice Chatbot

Accept voice input

4. Multi-language Support

Support different languages

5. Mobile Application

Android/iOS app

6. Database Integration

Replace JSON with database

REFERENCES

- Groq API Documentation

Official documentation of the Groq API used for integrating high-performance AI inference in the chatbot system. It provides details about API usage, request handling, authentication, and response generation.

Source: https://console.groq.com/docs

- LLaMA AI Model Documentation

Documentation related to the LLaMA (Large Language Model Meta AI) used for generating intelligent chatbot responses. It explains model architecture, capabilities, and implementation methods.

Source: https://ai.meta.com/llama/

- Flask Official Documentation

The official guide for the Flask web framework used to build the backend of the chatbot application. It includes routing, templates, request handling, and server configuration.

Source: https://flask.palletsprojects.com/

- Python Documentation

Official Python documentation used for understanding core programming concepts, libraries, and modules required in the chatbot development.

Source: https://docs.python.org/3/

- Research Papers on Chatbots

Various academic research papers and journals referred to for understanding chatbot design, natural language processing, and AI-based interaction systems. These sources provide theoretical background and advanced concepts.

- Online Tutorials and Articles

Various online learning platforms, blogs, and tutorials referred to during development for practical implementation guidance, debugging, and best practices. Examples include platforms like W3Schools, GeeksforGeeks, and Medium.

CHAPTER 10: APPENDICES

Appendix A: Sample JSON Data

```
{
  "hotel_name": "Royal Horizon Hotel",
  "location": "Parbhani, Maharashtra",
  "rooms": ["Standard", "Deluxe", "Suite"],
  "facilities": ["Wi-Fi", "AC", "Parking", "Restaurant"],
  "check_in": "2:00 PM",
  "check_out": "12:00 PM"
}
```

Appendix B: Sample Code

Flask Route Example:

```
@app.route('/chat', methods=['POST'])
def chat():
    user_message = request.form['message']
    response = get_ai_response(user_message)
    return jsonify({'response': response})
```

Clear Chat Function:

```
@app.route('/clear', methods=['POST'])
```

```
def clear():
    global chat_history
    chat_history.clear()
    return redirect('/')
```

CHAPTER 11: SOURCE CODE & EXECUTION

12.1 GitHub Link:

https://github.com/harshalirathod1105-rgb/hotel_chatbot

12.2 Steps to Run the Project

Step 1: Install Python

Download and install Python

Step 2: clone git repository from above provided link

Step 3: Install Required Libraries

(flask, groq ,virtual environment, etc.)

Step 4: Run the Application

1. Activate environment
2. Run: python app.py

Step 5: Open Browser

http://127.0.0.1:8000

12.3 Folder Structure

Project/

│— env/

```
│── templates/
        │── index.html
│── static/
        │── style.css
│── hotel_info.json
│── app.py
```

... THANK- YOU...